

Courier Management System (CMS)

Rephrased Project Document

Courier administration solution (CMS)

User Story:

Courier and logistics companies frequently face difficulties in effectively handling shipments and deliveries because they depend on manual procedures and separate systems. The lack of a centralised shipment tracking facility makes real-time monitoring of parcel movement difficult, resulting in delayed delivery updates and poor visibility for

customers. Delivery agents are assigned through manual procedures, which raises the possibility of scheduling conflicts and operational delays. Furthermore, the absence of automatic notifications makes timely customer communication difficult, while limited performance analytics reduce administration visibility into operational efficiency.

Handling several branches becomes difficult without an integrated platform, which can configure coordination problems between Field Locations. Together, These problems result in lower customer satisfaction, decreased productivity, and inefficient operations, and and may ultimately reduce organizational revenue.

1. INTRODUCTION

1.1 Project Overview :

Courier and logistics companies encounter difficulties in effectively managing shipments and deliveries due to manual operations and disconnected systems. These problems result in delayed updates, poor customer visibility, and operational inefficiencies. The Courier administration solution (CMS) is a Salesforce-based cloud solution that

brings shipment booking, tracking, and delivery-agent assignment together, automatic status updates, billing, and analytical reporting. The solution enhances operational efficiency, enhances customer satisfaction, and offers real-time business particulars.

1.2 Purpose :

The main objective of the Courier administration solution (CMS) is to automate and bring together courier and logistics operations with the Salesforce platform. The solution is intended to replace manual and scattered procedures with a unified digital platform that supports effective shipment booking, real-time tracking, delivery agent assignment,

automatic notifications, billing, and performance reporting. By offering one integrated platform for managing shipments across multiple branches, CMS increases operational visibility, strengthens communication with customers, minimizes delays and mistakes, and supports quicker decision-making through analytical insights. Ultimately, the solution is

designed to enhance service efficiency, customer satisfaction, and overall business returns.

2. IDEATION PHASE

2.1 Issue Statement (Customer Issue Statement Template)

PS I am (Customer) I'm trying

to But Because Which

makes me feel

PS-1

Branch Manager handle shipments

effectively tracking is manual systems are

disconnected stressed

PS-2

Customer track parcels updates are

delayed no centralised solution

uncertain

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2.2 Empathy Map Framework

User Persona: Branch Manager

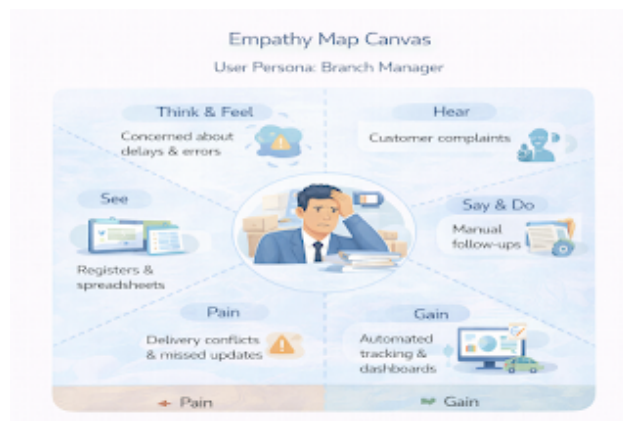
- Think & Feel: Concerned about delays & errors
- Hear: Customer complaints
- See: Registers & spreadsheets
- Say & Do: Manual follow-ups
- Pain: Delivery conflicts & missed updates
- Gain: automatic tracking & dashboards

2.3 Idea Generation & Priority Analysis

Ideas Reviewed

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- Manual registers
- Spreadsheet tracking
- Standalone portal
- Salesforce CRM

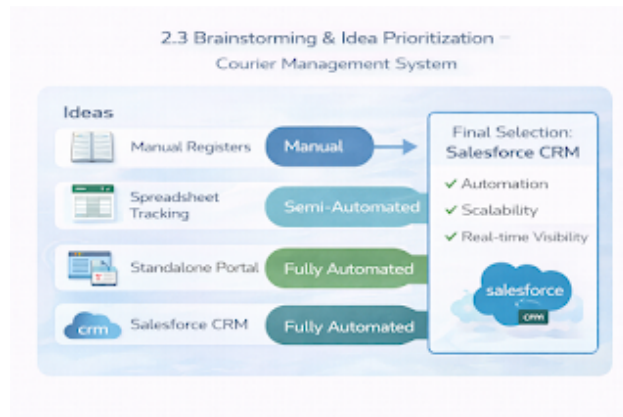
chosen Solution

Salesforce CRM chosen because of:

- Automation
- Scalability
- Live visibility

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3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Process Journey

Stages:

● Booking → Shipment Creation → Agent Assignment → Transit → Delivery

Update → Invoice → Reporting

3.2 needed Solution Features

Functional Needs FR No Requirement

FR-1 Customer administration

FR-2 Shipment booking

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FR-3 Delivery agent assignment

FR-4 Real-time status updates

FR-5 Invoice generation

FR-6 Reports & Dashboard Configuration Non-Functional Needs

NFR Description Usability Easy Lightning interface

Security Role-based access control Reliability Error-free automation

Performance Fast tracking updates Availability 24/7 cloud access

Scalability Supports growing branches

3.3 Data Flow Representation

Flow:

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports

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3.4 Technology Framework

Layer Technology

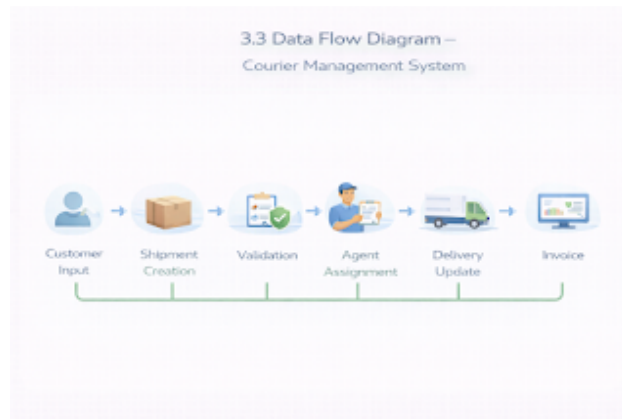
UI Salesforce Lightning Logic Salesforce Automation Flows

Database Salesforce Objects Automation Data Validation Rules, Automation Flows

Security Profiles, Roles Reporting Reports & Dashboard Configuration

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4. PROJECT DESIGN PHASE

4.1 Problem–Solution Alignment

CMS eliminates fragmented courier operations by automating shipment tracking, delivery assignment, billing, and reporting within a centralised Salesforce platform.

4.2 Suggested Solution

Functional Needs Parameter Description Problem

Manual shipment administration Solution Salesforce CMS Innovation

automatic tracking & billing

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Impact quicker deliveries Business Model Courier services

Scalability Multi-branch ready

4.3 Solution Structure

Components:

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Automation Flows
- Reports
- Dashboards

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5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning Approach

- Agile development method
- Epics → Stories → Story Points
- Execution through sprints
- Velocity measurement is utilized for estimation

Product Backlog, Sprint Schedule, and Estimation (4 Marks) Use the below template to configure product backlog and sprint schedule Sprint Functional

Requirement (Epic) User Story Number

User Story / Task Story Points Priority

Team Members Sprint-1 Developer Setup

USN-1 As a solution administrator, I want to configure and configure a Salesforce developer account so that I can set up and deploy the Courier administration solution. 3

High Member1

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Sprint Functional Requirement (Epic)

User Story Number User Story / Task Story

Points Priority Team Members

Sprint-2 Data Modeling

USN-2 As an admin, I want to configure Customer, Shipment, Delivery Agent, Branch, and Invoice objects with appropriate Object Relationships so that courier data is stored and managed effectively.

5 High Member2 Sprint-2

Data Modeling

USN-3 As a staff member, I want tabs and a Lightning App for all courier modules so that I can easily navigate and perform daily operations. 5 High

Member3 Sprint-3 Automation

USN-4 As a solution, I want to validate shipment particulars such as delivery dates, addresses, and mandatory fields so that incorrect or incomplete records are avoided.

3 High Member4 Sprint-3

Automation

USN-5 As a dispatcher, I want automatic Automation Flows for shipment status updates and agent assignment so that deliveries are processed quicker without manual intervention.

3 High Member4 Sprint-4

Security

USN-6 As an administrator, I want role-based access control for managers, agents, and staff so that sensitive shipment and billing data is secure. 5 High

Member2 Sprint-5 Reports

USN-7 As a branch manager, I want reports showing delivery performance, shipment status, and revenue so that I can track operations and make informed decisions. 4

High Member3 Sprint-6 Dashboards

USN-8 As a branch manager, I want dashboards showing delivery performance, shipment status, and revenue so that I can track operations and make informed decisions. 4

Medium Member4 Project Tracker, Velocity & Burndown Chart: (4 Marks) Sprint

Total Story Points Duration Sprint Start

Date Sprint End Date (Planned) Story Points

Completed (as on Planned End Date) Sprint Release

Date (Actual) Sprint-1 20 6 Days

24 Oct 2022 29 Oct 2022 20 29 Oct 2022

Sprint-2 20 6 Days 31 Oct 2022

05 Nov 2022 Sprint-3 20 6 Days

07 Nov 2022 12 Nov 2022

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Sprint Total Story Points Duration

Sprint Start Date Sprint End Date (Planned)

Story Points Completed (as on Planned End Date)

Sprint Release Date (Actual) Sprint-4 20

6 Days 14 Nov 2022 19 Nov 2022

6. PROJECT DEVELOPMENT STAGE:

Learning Outcomes

1. Practical Salesforce project implementation
2. Creating custom objects and their Object Relationships
3. Building Page Layout Configuration and record types
4. Defining Data Validation Rules
5. Creating automation with Automation Flows
6. Preparing reports and dashboards
7. Configuring security through profiles, permission sets, and sharing
8. Complete platform testing

Project Milestone 1: Salesforce Account

Introduction:

New to Salesforce and unsure where to start? Curious how real business applications—like loan processing and payment tracking—can be built on the platform? You're in the right place. Welcome to the Loan administration solution project!

Salesforce offers powerful tools that assist automate processes, enhance accuracy, and handle customer operations with ease. In this project, you'll learn how Salesforce can streamline lending workAutomation Flows—from borrower onboarding to loan approval, EMI tracking, and document verification.

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Salesforce Overview Salesforce is your customer success platform, designed to assist you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. With standard products and features, you can handle

Object Relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity 1: Developer Account Setup

Creating a developer org in Salesforce.

1. Go to @@PROT0@@
2. On the sign up form, Provide the below particulars :
3. First name & Last name
4. Email
- 1) First name & Last name
- 2) Email
- 3) Job Title: Developer
- 4) Company: College Name
- 5) Country: India

Click on sign me up after filling these.

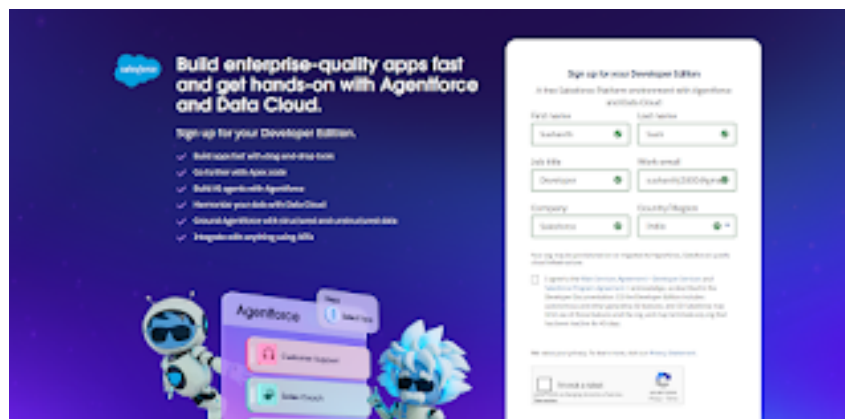
Activity 2: Activating the Account

1. Go to the inbox of the email that you utilized while signing up. Click on the verify

account to activate your account. The email may take 10-30 mins and sometimes 2

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hours.

2. Click on Verify Account

3. Give a password and answer a security question and click on change password.

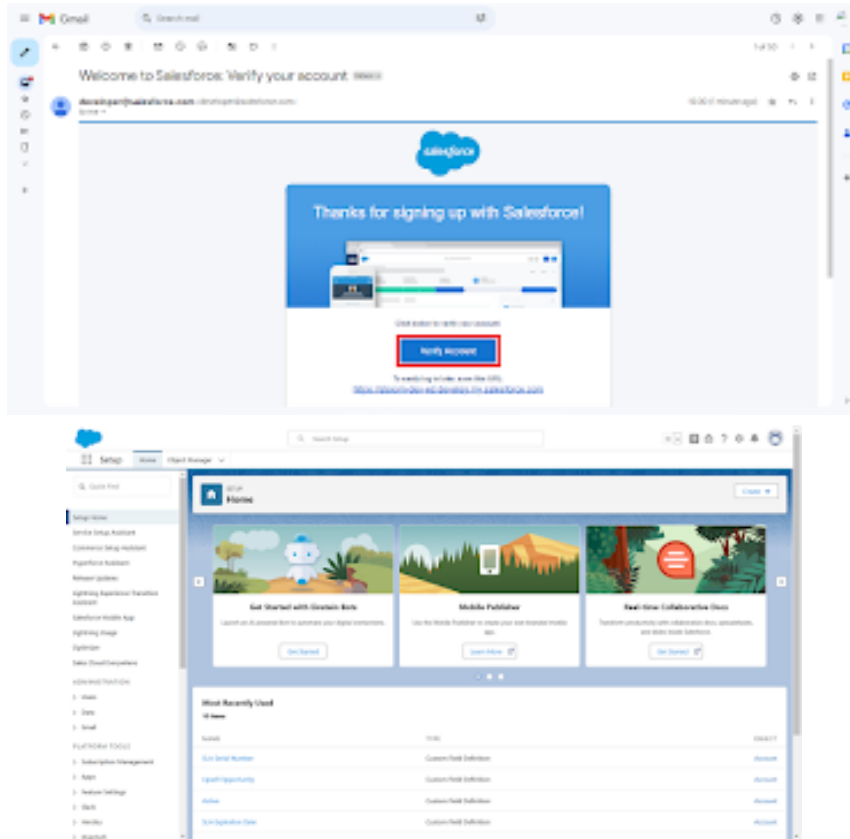
4. Then you will redirect to your salesforce setup page.

Project Milestone 2- Object:

Object Definition

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Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

Custom Objects Configured:

1. **@@PROT0@@** – stores customer personal and contact particulars
2. **@@PROT0@@** – manages parcel particulars, source, destination, and tracking particulars
3. **@@PROT0@@** – maintains delivery agent profiles and branch particulars
4. **@@PROT0@@** – represents courier branch offices
5. **@@PROT0@@** – stores status updates for each shipment
6. **@@PROT0@@** – manages billing particulars for delivered shipments

Object Relationships:

Customer → Shipment (Lookup) Shipment → Delivery_Agent (Lookup) Delivery_Agent → Branch (Lookup) Delivery_Update → Shipment (Master-Detail)

Invoice → Shipment (Lookup)

Implementation Procedure :

Activity 1: configure Customer Object

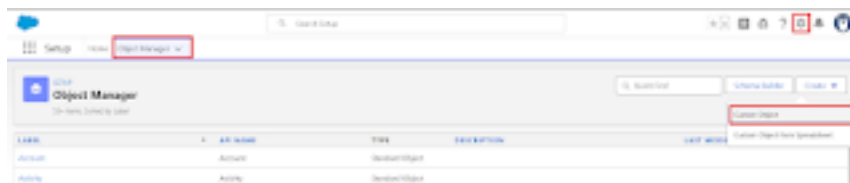
1. Click Setup → Object Manager → configure → Custom Object

2. Provide the below particulars:

● Label: Customer

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- Plural Label: Customers
- Record Name: Customer Name (Text)

3. Under Optional Features:

- Enable Allow Reports
- Enable Track Field History
- In Search Status check the box Allow Search
- In Object Creation Options check the box Add Notes and Attachments related list to default page layout

4. choose Save.

Activity 2: configure Shipment Object

1. Go to Setup → Object Manager → configure → Custom Object

2. Enter:

- Label: Shipment
- Plural Label: Shipments
- Record Name: Tracking Number (Auto Number)

3. Example Format: SHP-{00000}

Starting Number: 0

4. Enable:

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The screenshot shows the Salesforce Object Manager configuration page for a Custom Object. The 'Optional Features' section is highlighted with a red box, showing 'Allow Reports' and 'Track Field History' checked. The 'Object Classification' section shows 'Allow Reports' checked. The 'Deployment Status' section shows 'In Development' selected. The 'Search Status' section shows 'Allow Search' checked. The 'Object Creation Options' section shows 'Add Notes and Attachments' checked. The 'Save' button is highlighted with a red box.

- Allow Reports
- Track Field History

5. choose Save

Activity 3: configure Delivery_Agent Object

1. Setup → Object Manager → configure → Custom Object

2. Enter:

- Label: Delivery Agent
- Plural Label: Delivery Agents
- Record Name: Agent Name (Text)

3. Enable:

- Allow Reports
- Track Field History

4. choose Save

Activity 4: configure Branch Object

1. Setup → Object Manager → configure → Custom Object

2. Enter:

- Label: Branch
- Plural Label: Branches
- Record Name: Branch Name (Text)

3. Enable:

- Allow Reports

4. choose Save.

Activity 5: configure Delivery_Update Object

1. Setup → Object Manager → configure → Custom Object

2. Enter:

- Label: Delivery Update
- Plural Label: Delivery Updates
- Record Name: Update Number (Auto Number)

3. Example Format: UPD-{0000}

Enter Starting number : 0

4. Enable:

- Allow Reports
- Track Field History

NO THANKS

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5. choose Save.

Activity 6: configure Invoice Object

1. Setup → Object Manager → configure → Custom Object

2. Enter:

- Label: Invoice
- Plural Label: Invoices
- Record Name: Invoice Number (Auto Number)

3. Example Format: INV-{0000}

Enter Starting number: 0

4. Enable:

- Allow Reports
- Track Field History

5. choose Save

Project Milestone 3 - Tabs

Tab:

Tabs in Salesforce assist users view the particulars at a glance. It displays the data of objects and other web content in the platform.

Activity 1: Creating a Tab for Customer Object

1. Click Home, type Tabs in Quick Find, and choose Tabs.

2. Under Custom Object Tabs, choose New.

3. For Object, choose Customer.

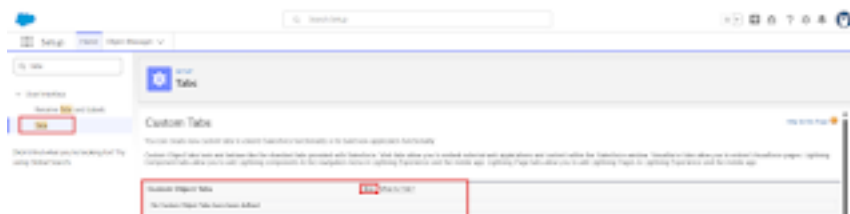
4. For Tab Style, choose any icon.

5. Keep the default settings unchanged.

6. choose Next, Next, and Save.

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Activity 2: Creating a Tab for Shipment Object

- 1. Click Home, type Tabs in Quick Find, and choose Tabs.**
- 2. Under Custom Object Tabs, choose New.**
- 3. For Object, choose Shipment.**
- 4. choose any tab style.**
- 5. Retain the default settings.**
- 6. choose Next, Next, and Save.**

Activity 3: Creating a Tab for Delivery Agent Object

- 1. Click Home, type Tabs in Quick Find, and choose Tabs.**
- 2. Under Custom Object Tabs, choose New.**
- 3. For Object, choose Delivery Agent.**
- 4. Pick any Tab Style.**
- 5. Leave default settings.**
- 6. choose Next, Next, and Save.**

Activity 4: Creating a Tab for Branch Object

- 1. Click Home, type Tabs in Quick Find, and choose Tabs.**
- 2. Under Custom Object Tabs, choose New.**
- 3. For Object, choose Branch.**
- 4. choose any tab style.**

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5. Retain all default options.
6. choose Next, Next, and Save.

Activity 5: Creating a Tab for Delivery Update Object

1. Click Home, type Tabs in Quick Find, and choose Tabs.
2. Under Custom Object Tabs, choose New.
3. For Object, choose Delivery Update.
4. choose any tab style.
5. Keep default configurations.
6. choose Next, Next, and Save.

Activity 6: Creating a Tab for Invoice Object

1. Click Home, type Tabs in Quick Find, and choose Tabs.
2. Under Custom Object Tabs, choose New.
3. For Object, choose Invoice.
4. choose any tab style.
5. Leave defaults unchanged.
6. choose Next, Next, and Save.

Project Milestone 4: Fields & Object Relationships

Field Definition Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are needed for a particular object in a record.

NO THANKS

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Activity 1: Field Configuration for the Customer Object

Field Name API Name Data Type needed

Description Customer Name Name

Text Yes Full name of customer Email

@@PROT0@@ Email Yes Customer email

Phone @@PROT0@@ Phone Yes

Customer contact number Address @@PROT0@@

Text Area Yes Customer address Customer Type

@@PROT0@@ Picklist Yes Regular / Business

City @@PROT0@@ Text No

Customer city State @@PROT0@@ Text

No Customer state Pincode @@PROT0@@

Text No Postal code Is Active

@@PROT0@@ Checkbox No Active status

1. Email (Email Field)

- Go to Setup → Object Manager → @@PROT0@@
- Click Fields & Object Relationships → New
- choose Email → Next
- Field Label: Email
- Check the needed box
- choose Next → Next → Save

2. Phone (Phone Field)

- Setup → Object Manager → @@PROT0@@
- Fields & Object Relationships → New
- choose Phone → Next

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- Field Label: Phone
- Check needed
- choose Next → Next → Save

3. Address (Text Area)

- Setup → Object Manager → @@PROTO@@
- Fields & Object Relationships → New
- choose Textarea → Next
- Field Label: Address
- Check needed
- Save

4. Customer Type (Picklist)

- Setup → @@PROTO@@ → Fields → New
- Picklist → Next
- Field Label: Customer Type
- **Enter values:**
 - Regular
 - Business
- Check needed
- Save

5. City (Text)

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- Setup → @@PROT0@@ → Fields → New
- Text → Next
- Field Label: City
- Save

6. State (Text)

- Setup → @@PROT0@@ → Fields → New
- Text → Next
- Field Label: State
- Save

7. Pincode (Text)

- Setup → @@PROT0@@ → Fields → New
- Text → Next
- Field Label: Pincode
- Save

8. Is Active (Checkbox)

- Setup → @@PROT0@@ → Fields → New
- Checkbox → Next
- Field Label: Is Active
- Save

NO THANKS

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Activity 2: Field Configuration for the Shipment Object

Field Name API Name Data Type

needed Description Tracking Number

Name Auto Number Yes Unique ID

Customer @@PROT0@@ Lookup(Customer) Yes

Related customer Source Address @@PROT0@@

Text Area Yes Pickup Field Location Destination

Address @@PROT0@@ Text Area Yes

Delivery Field Location Weight @@PROT0@@ Number

Yes Parcel weight Price @@PROT0@@

Currency Yes Shipping charge Status

@@PROT0@@ Picklist Yes Shipment status

Expected Date @@PROT0@@ Date

Yes Delivery estimate Assigned Agent

@@PROT0@@ Lookup(Delivery_Agent) No Delivery agent

Shipment Type @@PROT0@@ Picklist

Yes Domestic/Internation

1. Source Address (Text Area)

● Setup → @@PROT0@@ → Fields → New

● Textarea → Next

● Field Label: Source Address

● Mark as needed → Save

2. Destination Address (Text Area)

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- Setup → @@PROT0@@ → Fields → New
- Textarea → Next
- Field Label: Destination Address
- Mark as needed → Save

3. Weight (Number)

- Setup → @@PROT0@@ → Fields → New
- Number → Next
- Field Label: Weight
- Mark as needed → Save

4. Price (Currency)

- Setup → @@PROT0@@ → Fields → New
- Currency → Next
- Field Label: Price
- Mark as needed → Save

5. Status (Picklist)

Values:

- Booked
- In Transit
- Out for Delivery

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- Delivered

- Cancelled

Mark as needed → Save

6. Expected Date (Date)

- Setup → @@PROT0@@ → Fields → New

- Date → Next

- Field Label: Expected Date

- Mark as needed → Save

7. Shipment Type (Picklist)

Values:

- Domestic

- International

Mark as needed → Save

Activity 3: Field Configuration for Delivery_Agent Object

Field Name API Name Data Type needed

Agent Name Name Text Yes

Email @@PROT0@@ Email Yes

Phone @@PROT0@@ Phone Yes

Branch @@PROT0@@ Lookup Yes

Status @@PROT0@@ Picklist Yes

Joining Date @@PROT0@@ Date No

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1. Email

- Setup → @@PROT0@@ → Fields → New
- Email → Next
- Field Label: Email
- Mark as needed → Save

2. Phone

- Setup → @@PROT0@@ → Fields → New
- Phone → Next
- Field Label: Phone
- Mark as needed → Save

3. Status (Picklist)

Values:

- Active
- Inactive
- On Leave

Mark as needed → Save

4. Joining Date

- Setup → @@PROT0@@ → Fields → New
- Date → Next

NO THANKS

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- Field Label: Joining Date

- Save

Activity 4: Field Configuration for Branch Object

Field Name API Name Data Type needed

Branch Name Name Text Yes

Branch Code @@PROT0@@ Text Yes

City @@PROT0@@ Text Yes

State @@PROT0@@ Text Yes

Manager Name @@PROT0@@ Text

No

configure each field with:

- Setup → @@PROT0@@ → Fields → New

- choose respective Data Type

- Enter Field Label

- Save

(Repeat for all fields)

Activity 5: Field Configuration for Delivery_Update Object

Field Name API Name Data Type needed

Update Number Name Auto Number

Yes Shipment @@PROT0@@ Master-Detail

Yes Status @@PROT0@@ Picklist

Yes Update Date @@PROT0@@ DateTime

Yes Remarks @@PROT0@@ Text

No

NO THANKS

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Status Picklist Values:

- In Transit
- Out for Delivery
- Delivered
- Delayed

Mark as needed → Save

Activity 6: Field Configuration for Invoice Object

Field Name API Name Data Type needed

Invoice Number Name Auto

Number Yes Shipment @@PROT0@@

Lookup Yes Amount @@PROT0@@

Currency Yes Invoice Date @@PROT0@@

Date Yes Payment Status @@PROT0@@

Picklist Yes Payment Status Values

- Paid
- Unpaid
- Pending

Mark as needed → Save

Object Relationships :

Child Object Field Parent Object Type

Shipment @@PROT0@@ Customer Lookup

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Shipment @@PROT0@@ Delivery_Agent Lookup

Delivery_Agent @@PROT0@@ Branch Lookup

Delivery_Update @@PROT0@@ Shipment Master-Detail

Invoice @@PROT0@@ Shipment Lookup

Procedure for Creating Relationships :

Activity 1 – Shipment → Customer

- Setup → @@PROT0@@ → Fields → New
- Lookup Relationship
- Related Object: @@PROT0@@
- Field Label: Customer
- Save

Activity 2 – Shipment → Delivery Agent

- Setup → @@PROT0@@
- New → Lookup
- Object: @@PROT0@@
- Label: Assigned Agent
- Save

Activity 3 – Delivery Agent → Branch

- Setup → @@PROT0@@
- New → Lookup

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- Related Object: @@PROT0@@
- Label: Branch
- Save

Activity 4 – Delivery Update → Shipment

- Setup → @@PROT0@@
- New → Master-Detail
- Related Object: @@PROT0@@
- Save

Activity 5 – Invoice → Shipment

- Setup → @@PROT0@@
- New → Lookup
- Object: @@PROT0@@
- Save

Project Milestone 5-Data Validation Rules :

Data Validation Rules are applied when a user tries to save a record and are utilized to check if the data meets specified criteria. If the criteria are not met, the validation rule triggers an Error Text and prevents the user from saving the record until the problems are resolved.

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Activity 1: configure Data Validation Rules on @@PROT0@@

1. Validation Rule: Email Must Not Be Blank

Steps:

- Click the Gear icon → Setup
- Go to Object Manager
- Search and choose @@PROT0@@
- From the left menu, click Data Validation Rules
- choose New
- Rule Name: Email_Required
- Check Active

In Error Condition Formula, paste:

ISBLANK(@@PROT0@@)

- Click Check Syntax
- Error Text: Email is needed.
- Error Field Location: Field → Email
- choose Save.

2. Validation Rule: Phone Must Be 10 Digits

Steps:

- Setup → Object Manager → @@PROT0@@
- Data Validation Rules → New

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- Rule Name: Validate_Phone_Number

- Check Active

Formula:

LEN(@@PROTO@@) <> 10

- Check Syntax

- Error Text: Phone number must be 10 digits.

- Error Field Location: Field → Phone

- Save

3. Validation Rule: Pincode Must Be 6 Digits

Steps:

- Setup → Object Manager → @@PROTO@@

- Data Validation Rules → New

- Rule Name: Validate_Pincode

- Active → Yes

Formula:

LEN(@@PROTO@@) <> 6

- Error Text: Pincode must be 6 digits.

- Field Location: Field → Pincode

- Save

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Activity 2: configure Data Validation Rules on @@PROT0@@

1. Validation Rule: Weight Must Be Greater Than Zero

Steps:

- Setup → Object Manager → @@PROT0@@
- Click Data Validation Rules → New
- Rule Name: Validate_Weight
- Active → Checked

Formula:

@@PROT0@@ <= 0

- Check Syntax
- Error Text: Weight must be greater than zero.
- Field Location: Field → Weight
- Save

2. Validation Rule: Price Must Be Greater Than Zero

Steps:

- Setup → Object Manager → @@PROT0@@
- Data Validation Rules → New
- Rule Name: Validate_Price
- Active → Yes

Formula:

@@PROT0@@ <= 0

- Error Text: Price must be greater than zero.
- Field Location: Field → Price
- Save

3. Validation Rule: Expected Date Cannot Be in the Past

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Steps:

- Setup → Object Manager → @@PROT0@@
- Data Validation Rules → New
- Rule Name: Validate_Expected_Date

Formula:

@@PROT0@@ < TODAY()

- Check Syntax
- Error: Expected delivery date cannot be in the past.
- Field Location: Field → Expected Date
- Save

Activity 3: configure Data Validation Rules on @@PROT0@@**1. Validation Rule: Email Must Be Valid****Steps:**

- Setup → Object Manager → @@PROT0@@
- Data Validation Rules → New
- Rule Name: Validate_Agent_Email
- Active → Yes

Formula:

NOT(CONTAINS(@@PROT0@@, "@"))

- Error Text: Enter a valid email address.

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- Field Location: Field → Email

- Save

Activity 4: configure Data Validation Rules on @@PROT0@@

1. Validation Rule: Branch Code Must Not Be Blank

Steps:

- Setup → Object Manager → @@PROT0@@

- Data Validation Rules → New

- Rule Name: Branch_Code_Required

Formula:

ISBLANK(@@PROT0@@)

- Error Text: Branch Code is mandatory.

- Field Location: Field → Branch Code

- Save

Activity 5: configure Data Validation Rules on @@PROT0@@

1. Validation Rule: Remarks needed If Status is Delayed

Steps:

- Setup → Object Manager → @@PROT0@@

- Data Validation Rules → New

- Rule Name: Remarks_Required_For_Delay

Formula:

AND(ISPICKVAL(@@PROT0@@, "Delayed"), ISBLANK(@@PROT1@@))

NO THANKS

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- Error Text: Remarks are needed when status is Delayed.
- Field Location: Field → Remarks
- Save

Project Milestone 6 - Page Layout Configuration :

Page Layout Configuration in Salesforce control what a user sees on a record page and how particulars is organized. They assist structure fields, sections, buttons, related lists, and other components in a clean and user-friendly way.

By customizing Page Layout Configuration, we make sure that users can easily enter, update, and view the needed data.

Activity 1: Edit Page Layout for @@PROT0@@

Steps:

- Setup → Object Manager → @@PROT0@@
- Click Page Layout Configuration
- Click the default layout (usually Customer Layout)

Sections to Add & Arrange:

- Open the default layout
- In the top palette, click "Section"
- Drag the Section and drop it anywhere on the layout

A pop-up appears → Fill particulars:

- Section Name: "Customer particulars"

● Layout:

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- 1-Column or
- 2-Column (recommended)
- Tab-Key Order: Top-Down
- Click OK

The new section will appear on the layout.

- Drop fields inside the section (it will highlight).
- Repeat the same for each field.

Section 1: Customer particulars

- Customer Name
- Customer Type
- Email
- Phone
- Is Active

Section 2: Address particulars

- Address
- City
- State
- Pincode

9. Edit the layout as needed

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10. choose Save.

Activity 2: Edit Page Layout for @@PROT0@@

Steps:

- Setup → Object Manager → @@PROT0@@
- Click Page Layout Configuration
- Open the default Shipment Layout

Sections to Add:

Section 1: Shipment particulars

- Tracking Number
- Customer
- Shipment Type
- Weight
- Price
- Status

Section 2: Address particulars

- Source Address
- Destination Address

NO THANKS

USE THE APP

Section 3: Delivery particulars

- Expected Date
- Assigned Agent
- Arrange fields properly
- choose Save

Activity 3: Edit Page Layout for @@PROT0@@

Steps:

- Setup → Object Manager → @@PROT0@@
- Page Layout Configuration
- Open default Delivery Agent Layout

Arrange Fields:

Agent particulars

- Agent Name
- Email
- Phone
- Branch
- Status
- Joining Date

extra particulars

- Related Shipments

NO THANKS

USE THE APP

- Related Delivery Updates

- Edit → Save

Activity 4: Edit Page Layout for @@PROT0@@

Steps:

- Setup → Object Manager → @@PROT0@@

- Page Layout Configuration

- Open default Branch Layout

Sections:

Branch particulars

- Branch Name

- Branch Code

- City

- State

- Manager Name

- Arrange fields

- choose Save

NO THANKS

USE THE APP

Activity 5: Edit Page Layout for @@PROT0@@**Steps:**

- Setup → Object Manager → @@PROT0@@
- Page Layout Configuration
- Open default Delivery Update Layout

Sections:

Update particulars

- Update Number
- Shipment
- Status
- Update Date
- Remarks
- Arrange properly
- choose Save

Activity 6: Edit Page Layout for @@PROT0@@**Steps:**

- Setup → Object Manager → @@PROT0@@
- Page Layout Configuration
- Open default Invoice Layout

Sections:

NO THANKS

USE THE APP

Invoice particulars

- Invoice Number
- Shipment
- Amount
- Invoice Date
- Payment Status
- Arrange fields as needed
- choose Save

Project Milestone 7 - Automation Flows

In Salesforce, a flow is a powerful tool that enables you to automate business processes, collect and update data, and guide users through a series of screens or steps. Automation Flows are built with a visual interface and can be configured without any coding knowledge.

Activity 1: Auto-Update Shipment Status When Delivery

Update is configured

Scenario:

Whenever a @@PROT0@@ record is configured, automatically change the related @@PROT1@@ Status with the latest status from Delivery Update. Part 1: configure the Record-Triggered Flow Step 1: Open Flow Builder

- Click the Gear Icon → Click Setup
- In the Quick Find box (left side), type Automation Flows
- Click Automation Flows

NO THANKS

USE THE APP

- Click the New Flow button
- choose Record-Triggered Flow
- choose configure

Step 2: Configure the Start (Trigger)

On the Start panel on the right:

- Object: choose @@PROTO@@

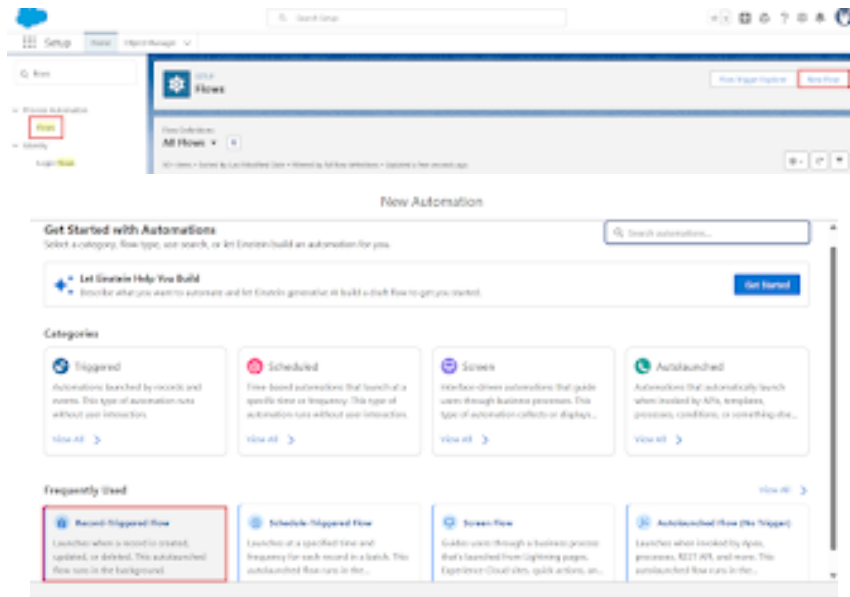
- **Trigger the Flow When:**

- choose A record is configured

- **Condition Requirements:**

NO THANKS

USE THE APP



■ Choose: None (Always Run the Flow)

● **Optimize the Flow For:**

■ choose Actions and Related Records

● choose Done

Part 2: Update Related Shipment Record Step 3: Add “Update Records” Element

● Click the + icon under the Start element

● choose Update Records

Configure the Update Records Element:

● Label: Update_Shipment_Status

● **How to Find Records to Update:**

■ choose Specify conditions to identify records

Set Filter Conditions:

● Object: @@PROT0@@

● Field: Id

● Operator: Equals

● Value: {!\$Record.@@@PROT1@@@}

(This indicates update the Shipment linked to the Delivery Update)

NO THANKS

USE THE APP

Set Field Values to Update

Add field mapping:

Shipment Field Value Source @@PROT0@@

{!\$Record.@@@PROT1@@@} (This copies status from Delivery Update to Shipment)

- choose Done

Part 3: Save and Activate Step 4: Save Flow

- choose Save (top-right)
- Flow Label: Auto_Update_Shipment_Status
- API Name will auto-fill
- choose Save

Step 5: Activate Flow

- choose Activate
- Flow status should show Active

NO THANKS

USE THE APP

Final Flow Layout:**Project Milestone 8 - Lightning platform :**

App Definition Apps in Salesforce are a group of tabs that assist the platform function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1: configure a Lightning App

configure the Courier administration solution App

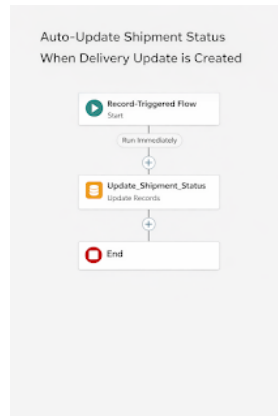
1. From Setup, enter App Manager in the Quick Find and choose App Manager.

2. choose New Lightning App.

■ Enter the Courier administration solution as the App Name.

NO THANKS

USE THE APP



3. Upload an image or icon for the app, then choose Next.

4. Under App Options, leave the default selections and choose Next.

5. Under Utility Items, leave as it is and choose Next.

6. From Available Items, choose the below and move them to chosen Items:

- Customer
- Shipment
- Delivery Agent
- Branch
- Delivery Update
- Invoice
- Reports

NO THANKS

USE THE APP

New Lightning App

App Details & Branding

Give your lightning app a name and description, upload an image and choose the highlight color for its navigation bar.

App Details

*** App Name** !

*** Developer Name** !

Description !

App Branding

Image !



[Clear](#)

Primary Color Hex Value !

■

Org Theme Options

☐ Use the app's image and color instead of the org's custom theme

[Cancel](#)

[Next](#)

■ Dashboards

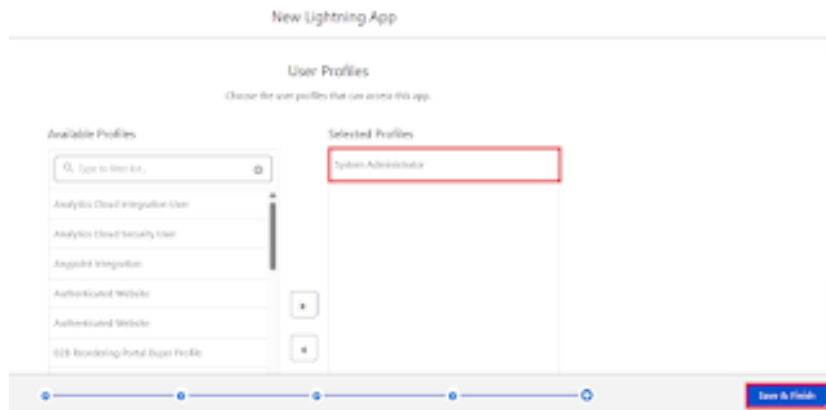
7. choose Next.**8. From Available Profiles, choose solution Administrator and move it to chosen Profiles.****9. choose Save & Finish.****Project Milestone 9- Profile:**

Profile Definition A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls "Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page Layout Configuration, Record Types,

Login hours & Login IP ranges. A profile can be assigned to many users, but users can be assigned a single profile at a time.

NO THANKS

USE THE APP



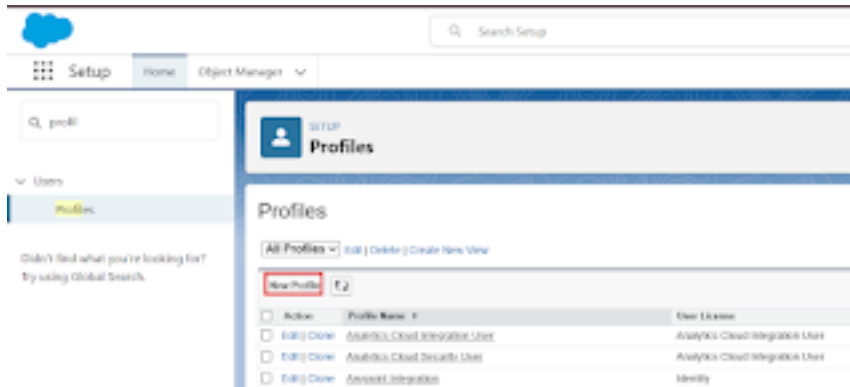
Activity 1: configure a Custom Profile – Courier Admin

configure a Custom Profile:

1. From Setup, enter Profiles in the Quick Find box.
2. Click Profiles.
3. choose New Profile.
4. choose Existing Profile as: solution Administrator.
5. Enter Profile Name: Courier Admin.
6. choose Save.

NO THANKS

USE THE APP



Why use solution Administrator as Base?

It already has:

- Modify All Data
- Customize platform
- Full CRUD access on all objects

By cloning it, the Courier Admin profile automatically inherits full access. You can later restrict permissions if needed.

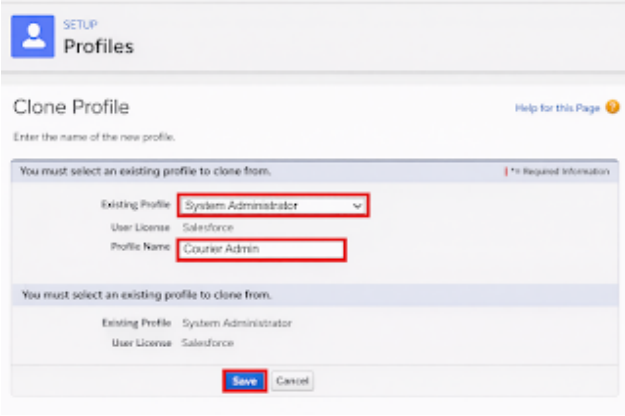
Activity 2: configure a Custom Profile – Branch Manager

Steps:

1. From Setup, enter Profiles in the Quick Find box.
2. Click Profiles.
3. choose New Profile.
4. choose Existing Profile as: Standard Platform User.
5. Enter Profile Name: Branch Manager.

NO THANKS

USE THE APP



SETUP
Profiles

Clone Profile [Help for this Page](#)

Enter the name of the new profile.

You must select an existing profile to clone from. * Required Information

Existing Profile	System Administrator
User License	Salesforce
Profile Name	Courier Admin

You must select an existing profile to clone from.

Existing Profile	System Administrator
User License	Salesforce

Save Cancel

6. choose Save.**7. Click Edit on the Branch Manager profile.**

Set App & Tab Settings

Under Assigned Apps / Custom App Settings, set the default app to:

Courier administration solution App

make sure Visible Tabs Include:

Set these tabs as Default On or Visible:

- Customers
- Shipments
- Delivery Agents
- Branches
- Delivery Updates
- Invoices
- Reports
- Dashboards

Custom Object Permissions

Scroll to Custom Object Permissions and set access as follows:

Object Permissions to Grant @@PROT0@@ configure, Read, Edit

NO THANKS

USE THE APP

@@PROT0@@ configure, Read, Edit @@PROT1@@ Read, Edit
@@PROT0@@ Read, Edit @@PROT1@@ configure, Read, Edit
@@PROT0@@ Read Session & Password Settings

Under Session Settings:

- Session Timeout: 2 Hours of Inactivity

Under Password Policies:

NO THANKS

USE THE APP

Custom Object Permissions							
	Basic Access				Data Administration		
	Read	Create	Edit	Delete	View All Records 	Modify All Records 	View All Fields
Customer__c	☒	☒	☒	☒	☐	☐	☐
Shipment__c	☒	☒	☒	☒	☐	☐	☐
Delivery_Agent__c	☒	☒	☒	☐	☐	☐	☐
Branch__c	☒	☒	☒	☐	☐	☐	☐
Delivery_Update__c	☒	☒	☒	☒	☐	☐	☐
Invoice__c	☒	☐					

- User passwords expire in: Never Expires
- Minimum Password Length: 8

choose Save.

Activity 3: configure a Custom Profile – Delivery Agent

Steps:

- 1. Setup → Quick Find → Profiles**
- 2. choose New Profile**
- 3. Existing Profile: Standard Platform User**
- 4. Profile Name: Delivery Agent**
- 5. choose Save**

App & Tab Access Default App: Courier administration solution

Visible Tabs:

- Shipments
- Delivery Updates

Object Permissions :

Object Permissions @@PROT0@@ Read, Edit

@@PROT0@@ configure, Read, Edit

NO THANKS

USE THE APP

@@PROT0@@ Read @@PROT1@@ Read

Restrictions

- No access to Branch editing
- No delete permissions
- Limited reporting access

choose Save.

Activity 4: configure a Custom Profile – Customer Support

Steps:

- Setup → Profiles → New Profile
- Base Profile: Standard User
- Profile Name: Customer Support

Access Configuration

Tabs Visible:

- Customers
- Shipments
- Delivery Updates

Object Permissions :

NO THANKS

USE THE APP

Object Access @@PROT0@@ Read, Edit

@@PROT0@@ Read, Edit @@PROT1@@ Read

@@PROT0@@ Read Session Settings

● Session Timeout: 2 Hours

● Password Length: 8

choose Save

Final Profiles configured in Project :

Profile Name Purpose Courier Admin Full solution control

Branch Manager handle shipments and agents Delivery Agent Update delivery status

Customer Support Handle customers and tracking

NO THANKS

USE THE APP

Project Milestone 10- Role:

In Salesforce, roles are utilized to determine which users have access to certain data and functions within the solution. They are also utilized to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the solution.

Activity 1: Creation of Courier Admin Role

Role Setup:

1. From the Quick Find box, search for Roles and click on the Roles option.
2. choose Set Up Roles.
3. Click on Expand All and click Add Role under CEO.

NO THANKS

USE THE APP



4. Give Label as “Courier Admin”.

■ The Role Name will auto-populate.

5. choose Save.

Activity 2: Creating Branch Manager Role Under Courier

Admin Role

configure another role:

1. configure a role with the name “Branch Manager”.

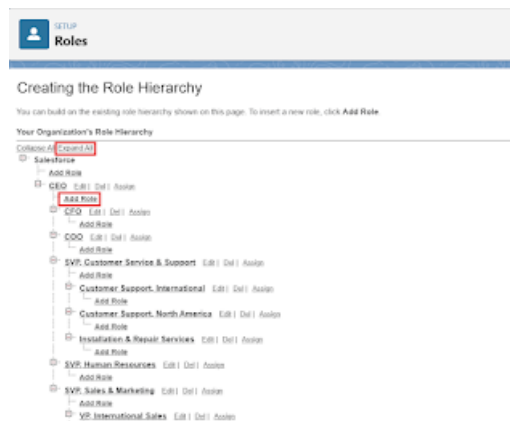
2. Below the Courier Admin, click on Add Role.

3. Enter the label name as “Branch Manager” and the Role Name will automatically populate.

4. choose Save.

NO THANKS

USE THE APP



Activity 3: Creating Delivery Agent Role Under Branch

Manager Role

Steps:

1. Below the Branch Manager role, click on Add Role.
2. Enter Label as “Delivery Agent”.

■ Role Name will auto-populate.

3. choose Save.

Activity 4: Creating Customer Support Role Under Branch

Manager

Steps:

1. Under the Branch Manager role, click on Add Role.
2. Enter Label as “Customer Support”.
3. Role Name will automatically populate.
4. choose Save.

Purpose of This Hierarchy :

- Courier Admin – Highest level access to all data
- Branch Manager – Manages branch-level operations
- Delivery Agent – Access only to assigned shipments
- Customer Support – Limited access to customer and shipment tracking

This hierarchy ensures appropriate data sharing and security in the Courier administration solution.

Project Milestone 11- User :

NO THANKS

USE THE APP

User Definition A user is anyone who logs in to Salesforce. Users are employees at your company, such as sales, reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account

settings determine what features and records the user can access. NOTE- As Salesforce license can only be utilized by 2 Users at a time in Dev Org, So If you don't find a salesforce license then deactivate a user who has a salesforce license Or change the license type from Salesforce to any other.

Activity 1: To configure a User

Steps:

1. From Setup, enter Users in the Quick Find box, then choose Users.

2. choose New User.

Complete the fields:

Field Name Value First Name Enter first name (Example: Ravi)

Last Name Enter last name (Example: Kumar) Alias Give an alias name

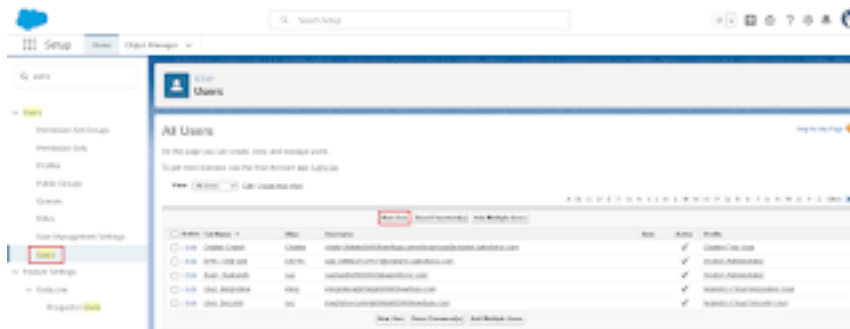
Email ID Enter your personal email ID Username Must be in the format: text@text.text and must be

unique Nickname Give a nickname Role

choose Courier Admin

NO THANKS

USE THE APP



User License Salesforce Profile Courier Admin

3. choose Save.

Activity 2: Creating Other Users

Follow the same procedure as Activity 1 for creating extra users.

Use the below particulars:

User Type Role User License Profile

Branch Manager Branch Manager Salesforce Branch Manager

Delivery Agent Delivery Agent Salesforce Platform

Delivery Agent Customer Support Customer

Support Salesforce Platform Customer

Support

Example:

To configure a Branch Manager User:

- Go to Setup → Users → New User

- **Fill particulars like:**

Field Example Value First Name Arjun

Last Name Reddy Email arjun@gmail.com

Username arjun.manager@cms.com Role Branch Manager

Profile Branch Manager

- choose Save

\

NO THANKS

USE THE APP

To configure Delivery Agent User:**Follow same steps with:**

- Role: Delivery Agent
- Profile: Delivery Agent

To configure Customer Support User:

- Role: Customer Support
- Profile: Customer Support

Important Points**● Every user must have:**

- Unique username
- Valid email
- Assigned Role
- Assigned Profile
- Profiles control permissions
- Roles control data visibility

Project Milestone 15 : Reports & Dashboard Configuration

Reports and Dashboards assist analyze business data and track courier operations effectively.

Activity 1: configure Report Types

Step 1: Open Report Types

1. Click Gear Icon → choose Setup.

NO THANKS

USE THE APP

2. In Quick Find search Report Types.

3. choose New Custom Report Type.

Step 2: configure Shipment Report Type

- Primary Object: @@PROT0@@
- Label: Shipment Tracking Report
- Category: Reports
- Deployment Status: Deployed
- choose Next → Save

Step 3: configure Customer Report Type

- Primary Object: @@PROT0@@
- Label: Customer Summary Report
- Category: Reports
- Deployment Status: Deployed
- choose Next → Save

Step 4: configure Agent Performance Report Type

- Primary Object: @@PROT0@@
- Label: Agent Performance Report
- Category: Reports
- Deployment Status: Deployed
- choose Next → Save

NO THANKS

USE THE APP

Activity 2: configure Reports

Report 1: Shipment Status Summary Report Step 1: configure the Report

1. Click App Launcher (9 dots) → Search Reports

2. choose New Report

3. Choose Shipment Tracking Report

4. Click Continue

Step 2: Add Columns

Add these fields:

- Tracking Number
- Customer
- Status
- Expected Date
- Assigned Agent

Step 3: Add Filters

- Show Me: All Shipments
- Date Filter: This Month
- Status Filter: choose specific Status (Optional)

Step 4: Group & Summarize

- Group rows by @@PROT0@@

NO THANKS

USE THE APP

- Add summary on Price

Step 5: Save Report

- choose Save & Run
- Name: Shipment Status Summary
- Folder: CMS Reports

Report 2: Agent Performance Report Step 1: configure Report

- New Report → Choose Agent Performance Report
- Click Continue

Step 2: Add Columns

- Agent Name
- Branch
- Number of Shipments
- Status

Step 3: Group Data

- Group by @@PROTO@@

Step 4: Save Report

- Name: Agent Performance Report
- Folder: CMS Reports
- Save & Run

NO THANKS

USE THE APP

Report 3: Revenue Report Steps

- configure report with Shipment Tracking Report Type

Add fields:

- Tracking Number
- Customer
- Price
- Invoice Status

Group by:

- Branch
- Month

Save as: Revenue by Branch Report

Activity 3: configure Dashboard

Step 1: Open Dashboards

- 1. Click App Launcher (9 dots)**
- 2. Search Dashboards**
- 3. choose New Dashboard**

Step 2: Enter Dashboard particulars

- Name: Courier administration Dashboard

NO THANKS

USE THE APP

- Folder: CMS Reports
- choose configure

Activity 4: Add Dashboard Components

Component 1: Shipment Status Overview (Donut Chart)

1. Click + Component
2. Choose report: Shipment Status Summary
3. Visual Type: Donut Chart
4. Click Add

Component 2: Agent Performance (Bar Chart)

- choose report: Agent Performance Report
- Chart Type: Bar Chart
- Click Add

Component 3: Revenue by Branch (Column Chart)

- Choose report: Revenue by Branch Report
- Chart Type: Column Chart
- Click Add

NO THANKS

USE THE APP

Activity 5: Save & Activate Dashboard

1. choose Save

2. choose Done

Share Dashboard With:

- Courier Admin
- Branch Managers
- Customer Support Team

Final Layout :

NO THANKS

USE THE APP

7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to verify record creation, validation rule enforcement, delivery status updates, flow automation, invoice generation, and dashboard visualization. All scenarios were validated to make sure accurate performance and reliability.

8. ADVANTAGES & DISADVANTAGES

Benefits

- centralised data
- automatic workAutomation Flows
- Real-time reporting

NO THANKS

USE THE APP



DisBenefits

- Salesforce dependency
- Requires initial configuration effort

9. Final Summary:

The Courier administration solution successfully digitizes courier operations with Salesforce automation, improving efficiency, accuracy, and customer satisfaction.

10. FUTURE SCOPE

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimization

11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

Thank You

NO THANKS

USE THE APP

NO THANKS
USE THE APP

